Sivanesan B

AI/ML Engineer | Generative AI | Forward Deployed Engineer (FDE)

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Profile Summary

AI/ML Engineer with hands-on experience in Machine Learning, Deep Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, and AI agent development. Currently working as a GenAI Executive & Forward Deployed Engineer, building AI-powered applications and business automation solutions. IEEE-published researcher in carbon-efficient LLM inference (58% CO_2 reduction, 56% faster inference), with experience in model optimization, computer vision, NLP, and end-to-end AI application development.

Experience

GenAI Executive & Forward Deployed Engineer (FDE) | *Nunes Instruments* **Aug 2026 – Present**

- Develop and integrate LLM-powered applications, AI agents, and Retrieval-Augmented Generation (RAG) workflows to automate business processes.
- Engineer AI-driven document processing, information extraction, and classification workflows using prompt engineering and REST API integration.
- Built a Flask-based WhatsApp business-intelligence dashboard that turns chat exports into searchable insights, and automated ERP-based Excel reporting in Python, replacing manual daily report preparation.

Innovation Engineer Trainee | *Forge Innovation & Ventures* **Jan 2025 – Jul 2025**

- Developed an offline voice-enabled AI inventory assistant using Whisper STT and a quantized local LLM, integrating FAISS semantic search with on-device inference for context-aware stock retrieval without internet.

Publications & Awards

Eco-Transformers: Carbon-Efficient and Accelerated Inference through Self-Pruning and Activation Freezing in LLMs **Aug 2025**

- Published at **IEEE ICNGCS 2025**. DOI: 10.1109/ICNGCS64900.2025.11183217.
- Achieved **58% CO_2 emission reduction** and **56% faster inference** in LLMs through self-pruning and activation freezing.
- **Best Final Year Project Award** – Department of Artificial Intelligence & Data Science, KCT.

Projects

NUNES AI Live Call Translator | Android APK

- Developed an AI-powered real-time Tamil ↔ Hindi voice call translation application using Kotlin, Plivo Voice API, and Sarvam AI for seamless multilingual communication.
- Applied Forward Deployed Engineering principles to design and implement a bidirectional speech processing pipeline (STT → Translation → TTS), integrating live transcripts, contact synchronization, and call history management.

Technologies Used: Kotlin, Jetpack Compose, Plivo, Sarvam AI, Node.js, TypeScript, Fastify, WebSocket, PostgreSQL.

AI Book Scanner APK

- Developed a Flutter-based AI book recognition app using fine-tuned ConvNeXt, achieving 92% detection accuracy across 1,000+ test images.
- Built a FastAPI inference service designed for 100+ concurrent users, integrating FAISS semantic search and VLM/LLM processing for intelligent book discovery.

Technologies Used: Flutter, Firebase, FastAPI, PyTorch, ConvNeXt, FAISS, VLM, LLM.

Technical Skills

Programming: Python, SQL, TypeScript
AI & Machine Learning: Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Agentic AI, LLMs, VLMs, Retrieval-Augmented Generation (RAG), AI Agents, Voice AI, LLM Evaluation
Forward Deployed Engineering: AI Solution Development, Business Requirements, System Integration, REST APIs, AI Workflow Automation, Model Deployment
AI Frameworks & Libraries: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, FastAPI, Flask, OpenAI API, Pandas, NumPy, scikit-learn
LLM Engineering: Prompt Engineering, Fine-tuning, LoRA, QLoRA, Embeddings, Model Optimization, Quantization, FAISS, ChromaDB, Semantic Search
Cloud & DevOps: Docker, Google Cloud, AWS, Firebase, Supabase, Hugging Face Hub, Git, GitHub
Certifications: IBM AI Engineering Professional Certificate, AWS AI & ML Scholars (Udacity)

Education

B.Tech in Artificial Intelligence and Data Science *(Lateral Entry)* **2023 – 2026**
Kumaraguru College of Technology, Coimbatore | CGPA: 8.02/10

Diploma in Computer Engineering **2020 – 2023**
Tamilnadu Government Polytechnic College, Madurai | CGPA: 8.0/10